A local-degree entropy bound for systematic Gibbs scans

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Abstract

Let $P_1, \dots, P_n$ be the single-site heat-bath operators of a finite probability law. Suppose that $\mathbf{Ent}_\mu(g^2) \le L \sum_i \|(I - P_i)g\|_2^2$ and that each update fails to commute with at most Δ others. We prove that every ordered sweep contracts entropy by a factor at most $L\Delta^2/(1 + L\Delta^2)$. Approximate tensorization with constant C and a lower bound b on positive full one-site conditional probabilities gives $L = C[2 + 2\log(1/b)]$. The estimate permits constrained support and prescribed orders that vary between sweeps. Only full one-site conditional probabilities are bounded below; no lower bound under arbitrary partial conditioning is required. The proof follows square roots of the evolving densities, bounds their terminal local residuals using commutation, and charges their squared movement to entropy loss. Applied to the established fully frustrated hypercube Ising model with $n = 2^d$ sites and fixed inverse temperature $0 < a < 1/2$, the estimate gives $O_a(d^{5/2}\log(n/\varepsilon))$ complete sweeps from every starting state, uniformly over prescribed update orders. We compare this conclusion with consequences of earlier proofs and with the faster bound available for a chosen checkerboard order.

Keywords: systematic scan, Gibbs sampling, entropy contraction, logarithmic Sobolev inequality, approximate tensorization, frustrated Ising model

Preprint, 11 October 2026. ORCID: 0009-0006-4105-1949

1 The local-degree comparison

A Gibbs sweep updates each coordinate once. After an update has equilibrated one coordinate conditionally on the others, later updates can disturb that equilibrium. The extent of this disturbance depends on which updates interact. We quantify this dependence through commutation and obtain an entropy estimate whose cost is quadratic in the maximum number of noncommuting neighbors.

In statistical mechanics, these are sequential heat-bath updates: the order of visiting the sites is prescribed, but the new spin at each site is sampled from its conditional Gibbs law. Each update preserves the joint Gibbs law, although a complete ordered sweep need not be reversible. The question is how uniformly the resulting stochastic procedure approaches equilibrium when the order is not chosen to suit the interactions. Section 4 interprets the entropy bound as decay of excess free energy for a frustrated spin model.

Blanca and Rafid [1] established a general systematic-scan entropy comparison from approximate tensorization, together with a quadratic-degree variance comparison. Their entropy bound has an exponential degree factor. Our argument retains the quadratic counting at the level of square roots of densities. The conditional square-root evolution is established in the sweeping-out approach to logarithmic Sobolev inequalities [2, 3], and the linear projection counting is established in the detectability lemma [4] and in [1]. The quantitative assertion developed here is the transfer from a supplied heat-bath logarithmic Sobolev inequality to the entropy of every ordered sweep.

Let μ be a probability measure with $\mu(x) > 0$ on a finite set

$$\Omega \subseteq \prod_{i=1}^{n} \Sigma_i.$$

We do not require Ω to be the full product. Write $\mu[h] = \sum_x \mu(x)h(x)$ and $\|h\|_2^2 = \mu[h^2]$. For $f \geq 0$, set

$$\mathrm{Ent}_\mu(f) = \mu[f \log f] - \mu[f] \log \mu[f],$$

with $0 \log 0 = 0$ and natural logarithms throughout. Let P_i be conditional expectation given all coordinates except i. It is an orthogonal projection on $L^2(\mu)$, preserves nonnegativity and mass, and is the single-site heat-bath operator.

A *commutation graph* has vertex set $\{1, \ldots, n\}$ and satisfies $P_i P_j = P_j P_i$ whenever $i \neq j$ are nonadjacent. Let its maximum degree be Δ. For an order $\pi = (i_1, \ldots, i_n)$, we use the density convention

$$Q_\pi = P_{i_n} \cdots P_{i_1}. \tag{1}$$

Thus P_{i_1} is applied first to the density. The Markov operator on observables for the same physical scan is Q_π^*; all orders are included in our statements.

Theorem 1.1 *Suppose that $L < \infty$, $L > 0$, and*

$$\mathrm{Ent}_\mu(g^2) \le L\mathcal{D}(g), \qquad \mathcal{D}(g) := \sum_{i=1}^{n} \|(I - P_i)g\|_2^2 \tag{2}$$

for every real function g on Ω. For every order π and every $f \ge 0$,

$$\mathrm{Ent}_\mu(Q_\pi f) \le \frac{L\Delta^2}{1 + L\Delta^2}\,\mathrm{Ent}_\mu(f). \tag{3}$$

When $\Delta = 0$, one sweep gives $Q_\pi f = \mu[f]$.

The normalization in (2) uses the sum of conditional variances, without division by n. In particular, L is a heat-bath logarithmic Sobolev constant. An inequality for an unweighted spin-flip gradient has a different Dirichlet form and cannot be substituted without a comparison of those forms.

The proof controls the local residuals of the final square-root density. This differs from applying a linear product of projections to $\sqrt{f}$: generally $\sqrt{P_i f} \ne P_i\sqrt{f}$. Earlier linear comparisons do yield an entropy bound on the Ln^2 scale after a standard conversion. Section 5 makes that implication explicit, so the comparison here concerns the local degree Δ, rather than claiming the first entropy consequence of such projection arguments.

2 Following the square-root density

For $g \ge 0$, define

$$T_i g = \sqrt{P_i(g^2)}, \qquad d_i(g) = \|(I - P_i)g\|_2 .$$

The map T_i follows the density update exactly and preserves its square-root norm. The quantity $d_i(g)$ is the distance from g to the subspace of functions independent of coordinate i. Immediately after applying T_i, this distance is zero. We first show that commuting updates cannot increase it.

Lemma 2.1 *For nonnegative g, h, $\left\|T_j g - T_j h\right\|_2 \le \|g - h\|_2$. If P_i and P_j commute, then $d_i(T_j g) \le d_i(g)$.*

Proof On each conditional fiber for P_j, the reverse triangle inequality for the weighted Euclidean norm gives

$$|\sqrt{P_j g^2} - \sqrt{P_j h^2}|^2 \le P_j[(g - h)^2].$$

Integration proves nonexpansivity. If $u \ge 0$ and $P_i u = u$, then u^2 is also P_i-measurable. Commutation implies that $P_j(u^2)$, and hence $T_j u$, is P_i-measurable. Since $P_i g \ge 0$, the distance to the range of P_i satisfies

$$d_i(T_j g) \le \left\|T_j g - T_j(P_i g)\right\|_2 \le \|g - P_i g\|_2 = d_i(g).$$

□



Fig. 1 Residual accounting in an illustrative sweep with order $1, \dots, 6$. Site 2 has neighbors $1, 3, 5$ in the commutation graph. Updating site 2 resets its residual to zero; only the later neighbors $3, 5$ can contribute to its terminal bound. Commuting updates at $4, 6$ cannot increase it. In the general proof, one factor of Δ bounds the number of later neighbors per site and another bounds how many earlier sites charge a given update. This is an accounting diagram, not a plot of computed residuals.

Lemma 2.2 *Let $g_k = T_{i_k} g_{k-1}$, $1 \le k \le n$, and $\delta_k = \|g_k - g_{k-1}\|_2$. Then*

$$\sum_i d_i(g_n)^2 \le \Delta^2 \sum_{k=1}^{n} \delta_k^2. \tag{4}$$

Proof Let $r(i)$ be the position of i in the order. We have $d_i(g_{r(i)}) = 0$. At a later nonneighbor update the residual cannot increase, by Lemma 2.1. At any other update, distance to a fixed closed subspace is a 1-Lipschitz function, so its increase is at most δ_k. Consequently

$$d_i(g_n) \le \sum_{\substack{k > r(i) \\ i_k \sim i}} \delta_k.$$

There are at most Δ terms. Cauchy–Schwarz followed by summation gives

$$\sum_i d_i(g_n)^2 \le \Delta \sum_i \sum_{\substack{k > r(i) \\ i_k \sim i}} \delta_k^2 \le \Delta^2 \sum_k \delta_k^2.$$

The second factor of Δ counts how many earlier neighbors can charge the same update. If $\Delta = 0$, all terminal residuals vanish and the same conclusion holds directly. □

Figure 1 isolates the chronological accounting in this proof. The bound uses which updates can increase a residual; it does not require the residual itself to increase monotonically between updates.

It remains to relate the accumulated movement to a quantity which decreases along the actual scan. Entropy has an exact loss identity for conditional expectation; its relative-entropy interpretation pays for the movement of the square roots.

Lemma 2.3 *If $f \ge 0$ and P is a conditional expectation under μ, then*

$$\left\| \sqrt{f} - \sqrt{Pf} \right\|_2^2 \le \mathrm{Ent}_\mu(f) - \mathrm{Ent}_\mu(Pf). \tag{5}$$

Proof The zero function is immediate; otherwise, by homogeneity, assume $\mu[f] = 1$. On a fiber where $Pf = 0$, f also vanishes. Elsewhere, measurability of $\log Pf$ gives

$$D(f\mu\|(Pf)\mu) = \mu[f\log(f/Pf)] = \mathrm{Ent}_\mu(f) - \mathrm{Ent}_\mu(Pf).$$

For two probability measures ν, η, relative entropy dominates squared Hellinger distance:

$$D(\nu\|\eta) \geq -2\log\sum_x \sqrt{\nu(x)\eta(x)} \geq \sum_x (\sqrt{\nu(x)} - \sqrt{\eta(x)})^2.$$

The first inequality is Jensen's inequality for $-\log$, and the second is $-\log u \geq 1-u$. Applying this to the preceding identity proves (5). Zero values can equivalently be handled by adding a positive constant and passing to the limit on the finite space. □

Proof of Theorem 1.1 Put $f_0 = f$, $f_k = P_{i_k} f_{k-1}$ and $g_k = \sqrt{f_k}$. Lemma 2.3 telescopes to

$$\sum_k \delta_k^2 \leq \mathrm{Ent}_\mu(f_0) - \mathrm{Ent}_\mu(f_n).$$

Use (2) at the terminal function g_n, and then Lemma 2.2, to obtain

$$\mathrm{Ent}_\mu(f_n) \leq L\sum_i d_i(g_n)^2 \leq L\Delta^2\big[\mathrm{Ent}_\mu(f_0) - \mathrm{Ent}_\mu(f_n)\big].$$

Rearrangement gives (3). For $\Delta = 0$, the entropy of $Q_\pi f$ is zero, so it is constant on the support; preservation of mass identifies that constant as $\mu[f]$. □

3 Tensorization and mixing consequences

Write $\mathrm{Ent}_i(f) = P_i(f\log f) - (P_i f)\log(P_i f)$ and $e_i(f) = \mu[\mathrm{Ent}_i(f)]$. Approximate tensorization with constant $0 < C < \infty$ means

$$\mathrm{Ent}_\mu(f) \leq C\sum_i e_i(f), \qquad f \geq 0. \tag{6}$$

To use Theorem 1.1, we convert this inequality to (2) on each full one-site conditional fiber.

Lemma 3.1 *Let ν be a finite probability measure whose positive atom weights are at least $b > 0$. Then*

$$\mathrm{Ent}_\nu(g^2) \leq [2 + 2\log(1/b)]\,\mathrm{Var}_\nu(g). \tag{7}$$

Proof First let $g \geq 0$ and $m = \nu[g] > 0$. Each $t = g(x)/m$ lies in $[0, 1/b]$. The function $\phi(t) = t^2\log(t^2) - t^2 + 1$ has $\phi(1) = \phi'(1) = 0$ and

$$\phi''(t) = 4 + 4\log t \leq 4 + 4\log(1/b), \qquad 0 < t \leq 1/b.$$

Taylor's integral formula, with continuity at zero, yields

$$\phi(t) \leq [2 + 2\log(1/b)](t-1)^2.$$

The variational identity $\mathrm{Ent}_\nu(h) = \inf_{c>0}\nu[h\log(h/c) - h + c]$, evaluated at $c = m^2$, now proves (7). If $m = 0$, $g = 0$. For real g, apply the nonnegative case to $|g|$ and use $\mathrm{Var}_\nu(|g|) \leq \mathrm{Var}_\nu(g)$. □

Corollary 3.2 *Assume* (6). *If every positive probability in every full one-site conditional law is at least $b > 0$, then* (3) *holds with*

$$A = C[2 + 2\log(1/b)]\Delta^2 \tag{8}$$

in place of $L\Delta^2$.

Proof Apply Lemma 3.1 on each fiber, average over the other coordinates, and use (6). This gives (2) with $L = C[2 + 2\log(1/b)]$. □

This corollary uses only the full one-site conditionals. It does not require connectedness of every partially conditioned support. The finite inequality (6) itself excludes nonconstant functions invariant under all updates: such a function would have every e_i zero while its entropy was positive.

Corollary 3.3 *Let $A = L\Delta^2 > 0$, let f be a probability density relative to μ, and let $Q_1, \dots, Q_t$ be sweeps in any externally prescribed sequence of orders. Then*

$$\|(Q_t \cdots Q_1 f)\mu - \mu\|_{\mathrm{TV}} \le \sqrt{\tfrac{1}{2}\operatorname{Ent}_\mu(f)}\, \exp\left(-\frac{t}{2(1+A)}\right).$$

If $|\Omega| > 1$, then for a point start it suffices for error at most ε that

$$t \ge (1 + A)\left[\log\frac{\log(1/\mu_{\min})}{2\varepsilon^2}\right]_+, \qquad \mu_{\min} = \min_{x\in\Omega}\mu(x), \tag{9}$$

rounded upward. Here $[u]_+ = \max\{u, 0\}$ and $\|\nu - \mu\|_{\mathrm{TV}} = \frac{1}{2}\sum_x |\nu(x) - \mu(x)|$. A singleton support is already at equilibrium at time zero.

Proof Iterate (3), use $A/(1 + A) \le e^{-1/(1+A)}$, and apply Pinsker's inequality. A point start at x has relative entropy $\log(1/\mu(x))$. □

The changing orders in this statement are prescribed independently of the current configuration; it asserts no bound for state-adaptive scheduling. If $A = 0$, one sweep suffices by Theorem 1.1. Full one-site lower bounds alone do not imply $\mu_{\min} \ge b^n$ on a constrained support, so (9) retains the minimum mass explicitly. For example, on the connected single-site path

$$\Omega = \{000, 001, 011, 111, 110\}, \qquad \mu \propto (1, r, r^2, r^3, r^4), \quad r = \frac{b}{1-b}, \quad 0 < b < \tfrac{1}{2},$$

every nondeterministic full conditional has probabilities $b, 1 - b$, whereas $b = 1/10$ gives $\mu_{\min} = 1/7381 < b^3$. Pinning the third coordinate to zero leaves the disconnected support $\{000, 110\}$. The full one-site lower bound thus implies neither the stated minimum-mass bound nor connectedness under every pinning.

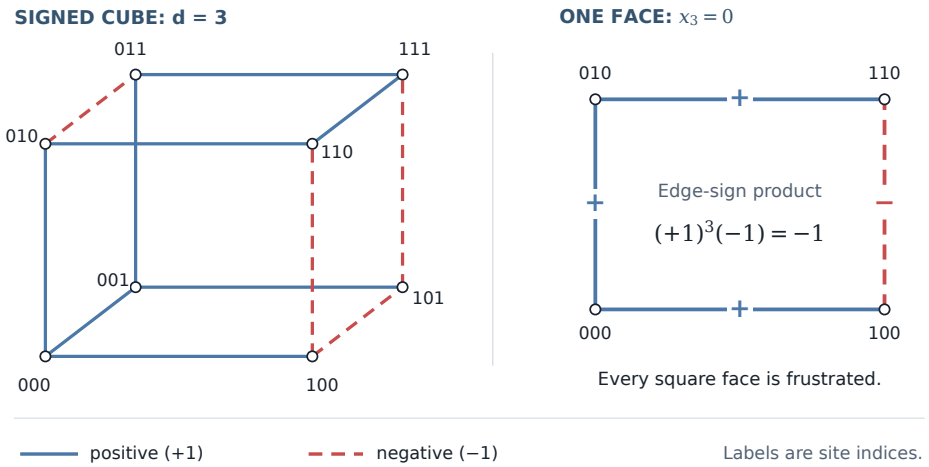


Fig. 2 The signed interaction graph for $d = 3$, with one square face displayed separately. Vertex labels are site coordinates x, not spin values σ_x. Positive and negative edges follow (10); their product around every square is -1. Thus no spin assignment can satisfy all four interactions on a square. The actual coupling on an edge is its sign times $a/\sqrt{d}$.

4 The fully frustrated hypercube

An application with $\Delta \ll n$ exposes the local nature of the bound. We use the established fully frustrated hypercube model described by Marinari, Parisi and Ritort [5] and Parisi and Potters [6]. Fix $d \geq 2$. There are $n = 2^d$ spin sites, indexed by $x \in \{0,1\}^d$, and each site has d neighbors. The coordinate x labels a site; the spin at that site is $\sigma_x \in \{-1, 1\}$.

The edge which flips coordinate j has sign

$$s_j(x) = (-1)^{x_1 + \cdots + x_{j-1}}. \tag{10}$$

Let U_j be the corresponding signed bit-flip matrix on the site space and $B_d = \sum_{j=1}^{d} U_j$. Since s_j does not depend on x_j, U_j is symmetric and $U_j^2 = I$. If $j < k$, flipping coordinate j changes s_k while flipping k does not change s_j. Hence $U_j U_k = -U_k U_j$, and

$$B_d^2 = dI. \tag{11}$$

This identity is also recorded in [6, §1.3, equation (9)]. Every square has negative edge-sign product. For instance, at $d = 2$ the horizontal edges are positive and the two vertical edges have opposite signs. Reversing any selection of spin labels preserves the sign product around a square, so these interactions cannot all be made ferromagnetic.

Fix $0 < a < 1/2$, and define

$$J_d = aI + \frac{a}{\sqrt{d}} B_d, \qquad \mu_d(\sigma) = Z_d^{-1} \exp\left(\tfrac{1}{2}\sigma^T J_d \sigma\right). \tag{12}$$

The diagonal term is constant in σ. The resulting off-diagonal interaction is the normalization in [5, equations (3)–(4)], with inverse temperature a. By (11), J_d is positive semidefinite with eigenvalues 0 and $2a$.

We use Lee's approximate tensorization theorem [7, Theorem 4.1]: for an Ising law proportional to $\exp(\sigma^T J\sigma/2 + h^T\sigma)$ with $J \succeq 0$ and $\|J\|_{\mathrm{op}} < 1$, (6) holds with $C \le (1 - \|J\|_{\mathrm{op}})^{-1}$, uniformly in h. Therefore $C \le (1-2a)^{-1}$ in (12). The largest absolute conditional field is $a\sqrt{d}$, and it is attained by aligning each neighbor with its incident edge sign. Full support makes this conditioning possible. The least one-site conditional atom is exactly

$$b_d = (1 + e^{2a\sqrt{d}})^{-1}.$$

Proposition 4.1 *For $d \ge 2$ and fixed $0 < a < 1/2$, every prescribed sweep of (12) satisfies (3) with $L\Delta^2$ replaced by*

$$A_d = \frac{2 + 2\log(1 + e^{2a\sqrt{d}})}{1 - 2a}\, d^2. \tag{13}$$

From every initial state, total-variation error at most $0 < \varepsilon < 1/2$ is achieved after

$$t \ge (1 + A_d) \log \frac{n(\log 2 + a)}{2\varepsilon^2} \tag{14}$$

complete sweeps, rounded upward. The asymptotic bound is

$$O_a\left(d^{5/2} \log(n/\varepsilon)\right).$$

At fixed error, this is $O_a((\log n)^{7/2})$.

Proof Nonadjacent site updates commute, so the interaction graph supplies $\Delta = d$. Corollary 3.2 gives (13). Also $0 \le \sigma^T J_d \sigma/2 \le an$, since $\|\sigma\|_2^2 = n$ and $0 \preceq J_d \preceq 2aI$. Thus $Z_d \le 2^n e^{an}$ and $\log(1/\mu_{\min}) \le n(\log 2 + a)$. Apply Corollary 3.3 and use $\log(1 + e^{2a\sqrt{d}}) \le 2a\sqrt{d} + \log 2$. □

Each sweep contains n single-site updates. This is a high-temperature guarantee uniform over prescribed orders. It makes no assertion about the glassy regime studied in [5], the optimal exponent, or the observed performance of a particular implementation.

4.1 Statistical-mechanical interpretation

The geometry here is a single d-dimensional hypercubic cell, with degree d and 2^d sites, as illustrated in Figure 2. Increasing d is different from taking larger boxes in a fixed-dimensional lattice. The distinction between the cell and the infinite hypercubic lattice is already emphasized in [6, §1.3].

Define the Hamiltonian without the irrelevant diagonal shift by

$$H_d(\sigma) = -\frac{1}{2\sqrt{d}} \sigma^T B_d \sigma.$$

Then $\mu_d(\sigma) \propto e^{-aH_d(\sigma)}$. In units with Boltzmann's constant equal to one, a is inverse temperature, and our range $0 < a < 1/2$ is $T > 2$ in this normalization. This

sufficient range is not asserted to be a sharp transition threshold. It lies well above the low-temperature regime investigated in [5].

For a probability law $\nu = f\mu_d$, write

$$S(\nu) = -\sum_{\sigma} \nu(\sigma) \log \nu(\sigma), \qquad F_a(\nu) = \nu[H_d] - a^{-1} S(\nu).$$

Substituting the Gibbs density in relative entropy gives the exact identity

$$\mathrm{Ent}_{\mu_d}(f) = D(\nu \| \mu_d) = a\big[F_a(\nu) - F_a(\mu_d)\big]. \tag{15}$$

Consequently, Proposition 4.1 bounds the decay of excess free energy per sweep for this stochastic update protocol. Relative entropy is the quantity that contracts; the Shannon entropy $S(\nu)$ alone need not increase at every update.

This finite-volume, worst-start guarantee addresses a different question from dynamical mean-field descriptions of sequential spin-glass updates. For example, Dandi, Gamarnik, Pernice and Zdeborová [8, Theorem 1 and Corollary 1] analyze finite-time empirical histories for random Sherrington–Kirkpatrick interactions through a large-system limit and a block approximation. That result does not supply the prescribed-order entropy estimate above. Conversely, our estimate does not describe aging, metastable states or critical relaxation in the low-temperature model.

5 Consequences of earlier comparisons

The value of Proposition 4.1 depends on comparing the same model, error criterion and time unit. We give the deductions used for that comparison, including stronger consequences than the displayed entropy theorem in [1].

5.1 A dimension-dependent entropy consequence

For a linear sweep Q, put $V_Q(g) = \|g\|_2^2 - \|Qg\|_2^2$. Before substituting a spectral gap, the proof of Gaitonde and Mossel [9, Theorem 3.1] gives

$$\mathcal{D}(g) \le 2n(n+1) V_Q(g).$$

Miclo's entropy-loss estimate [10, Lemma 5 and Proposition 6], with the density/observable adjoint convention aligned, gives

$$\mathrm{Ent}_{\mu}(f) - \mathrm{Ent}_{\mu}(Qf) \ge V_Q(\sqrt{f}).$$

Combining these two inequalities with (2) yields

$$\mathrm{Ent}_{\mu}(Qf) \le \left(1 - \frac{1}{2Ln(n+1)}\right) \mathrm{Ent}_{\mu}(f). \tag{16}$$

Kwon [11, §5.2, equation (42)] improves the comparison constant to $n^2 - n + 2$. These are deductions from the cited arguments; the Ln^2 entropy scale is already available. In

particular, Theorem 1.1 need not improve the asymptotic rate when Δ is proportional to n.

The distinction between input and output residuals prevents simply replacing n by Δ in the displayed linear inequality. For independent fair spins, parity $g(\sigma) = \prod_i \sigma_i$ satisfies $Qg = 0$, $\mathcal{D}(g) = n\,\|g\|_2^2$ and $V_Q(g) = \|g\|_2^2$, although every update commutes. Our proof instead uses residuals at the nonlinear terminal function $\sqrt{Qf}$.

5.2 Retaining the two-site entropy constant

The proof of [1, Theorem 4.4, equations (22)–(23)] yields a stronger model-specific comparison if its two-site constant is retained and one negative term is not discarded. Let P_{ij} condition outside $\{i, j\}$, and put $e_{ij}(f) = \mathrm{Ent}_\mu(f) - \mathrm{Ent}_\mu(P_{ij}f)$. Suppose every neighboring two-site conditional law satisfies approximate tensorization with constant at most $K \geq 1$. Then

$$\begin{aligned} e_i(P_j f) &\leq e_{ij}(P_j f) = e_{ij}(f) - e_j(f) \\ &\leq K e_i(f) + (K-1) e_j(f). \end{aligned} \tag{17}$$

The first inequality is conditional entropy decomposition, and the equality uses $P_{ij}P_j = P_{ij}$. Averaging the assumed pair inequality gives the last line.

For commuting updates and normalized f, data processing gives

$$e_i(P_j f) = D((P_j f)\mu \| (P_j P_i f)\mu) \leq D(f\mu \| (P_i f)\mu) = e_i(f).$$

Homogeneity extends this inequality to every $f \geq 0$. Along a sweep, the update at i sets its residual entropy to zero. Iterating (17) over at most Δ later neighbors therefore gives

$$e_i(Qf) \leq (K-1)K^{\Delta-1} \sum_{\substack{k > r(i) \\ i_k \sim i}} e_{i_k}(f_{k-1}), \qquad \Delta \geq 1.$$

Sum over i, apply (6), and telescope the update losses. This proves the prior-proof consequence

$$\mathrm{Ent}_\mu(Qf) \leq \frac{A_{\mathrm{pair}}}{1 + A_{\mathrm{pair}}} \mathrm{Ent}_\mu(f), \qquad A_{\mathrm{pair}} = C\Delta(K-1)K^{\Delta-1}. \tag{18}$$

It is the comparison used below, rather than only the coarser constant in the stated theorem of [1].

For an Ising pair with coupling w, arbitrary induced external fields are allowed. The harmless diagonal shift gives the interaction matrix

$$\begin{pmatrix} |w| & w \\ w & |w| \end{pmatrix},$$

whose eigenvalues are 0 and $2|w|$. Lee's theorem gives the uniform choice $K = (1-2|w|)^{-1}$ when $2|w| < 1$. In the hypercube model $|w| = a/\sqrt{d}$, so (18) permits

$$A_{\mathrm{pair}} = \frac{2a}{1-2a}\sqrt{d}\,(1-2a/\sqrt{d})^{-d} = O_a(\sqrt{d}\,e^{2a\sqrt{d}}). \tag{19}$$

For fixed $a > 0$, this chosen comparison parameter grows faster than every fixed power of d; (13) is polynomial. This compares proved guarantees, not lower bounds on the actual mixing time.

The pair route has a first-order edge cost even at zero external field. For $w \neq 0$, write $\rho = \tanh|w|$ and $h(x,y) = x + \mathrm{sgn}(w)y$. Then $\mathrm{Var}(h) = 2(1+\rho)$ and the sum of the two conditional variances is $2(1-\rho^2)$. Linearizing pair entropy tensorization at the constant density yields $K \geq (1-\rho)^{-1} = 1 + |w| + O(w^2)$. Thus replacing this particular pair constant by $1 + O(w^2)$ is not justified.

5.3 Faster sampling with a chosen order

Let E, O be the even and odd parity classes of the hypercube. Within each class, the spins are independent conditional on the other class. For a set S of sites, let P_S be conditional expectation given the spins outside S, and let $e_S(f) = \mathrm{Ent}_\mu(f) - \mathrm{Ent}_\mu(P_S f)$. Updating each spin of E once therefore implements P_E, in any order within E; the same holds for O. The physical checkerboard sweep, E followed by O, acts on densities as $P_O P_E$. Strong approximate Shearer factorization of Caputo, Chen and Parisi [12, Theorem 2.2] gives

$$\mathrm{Ent}_\mu(f) \leq C_{\mathrm{block}}[e_E(f) + e_O(f)], \qquad C_{\mathrm{block}} \leq (1-2a)^{-1}. \tag{20}$$

For their q-state normalization, encode the two spin values by $s = (1,-1)$ and take $\Gamma = J_d \otimes ss^T/2$. Its nonzero eigenvalues are those of J_d, and its Gibbs energy is (12). Choosing block weights $1/2, 1/2$ in their theorem gives (20).

Let $R = (P_E + P_O)/2$. Convexity of entropy and (20) imply

$$\mathrm{Ent}_\mu(Rf) \leq \left(1 - \frac{1}{2C_{\mathrm{block}}}\right)\mathrm{Ent}_\mu(f).$$

The absorption identity

$$P_O R P_E = P_O P_E$$

then gives the same contraction factor for the checkerboard sweep. Related layerwise projection identities are used by Guo, Kara and Zhang [13, proof of Theorem 1]. The resulting bound is $O_a(\log(n/\varepsilon))$ complete sweeps. Approximate tensorization also gives $O_a(n\log(n/\varepsilon))$ individual random-site updates, by applying entropy convexity to $n^{-1}\sum_i P_i$.

These faster guarantees concern schedules one can choose. Commuting swaps preserve the relative order of each interacting pair, so a general prescribed interleaving cannot be replaced by two consecutive parity blocks through commutation. The monotonicity and censoring assumptions used in the systematic-scan comparison of

Blanca and Zhang [14, Theorem 1.6 and §4.1] are not supplied by the frustrated interactions.

Table 1 Bounds on the same hypercube model, for fixed $0 < a < 1/2$. Time is measured in complete sweeps; the last row explicitly uses individual updates. All entries are upper bounds.

Comparison	Orders covered	Bound
Linear proof and entropy conversion, (16)	Every prescribed order	$O_a(n^2\sqrt{d}\log(n/\varepsilon))$
Retained pair estimate, (19)	Every prescribed order	$O_a(\sqrt{d}\,e^{2a\sqrt{d}}\log(n/\varepsilon))$
Proposition 4.1	Every prescribed order	$O_a(d^{5/2}\log(n/\varepsilon))$
Block factorization, (20)	Checkerboard	$O_a(\log(n/\varepsilon))$
Single-site tensorization	Random site	$O_a(n\log(n/\varepsilon))$ updates

Table 1 identifies the contribution of the local-degree estimate: it provides a polynomial-in-degree guarantee uniformly over prescribed orders on this connected frustrated family. The construction, entropy factorization, square-root evolution and projection counting are inherited ingredients. The chosen checkerboard order has the smaller bound established above.

Statements and Declarations

Funding

This research received no external funding.

Competing interests

The author declares no competing interests relevant to this work.

Data and code availability

The results are proved analytically and do not rely on empirical data or numerical simulation. The LaTeX source and the deterministic Python code used to draw Figures 1 and 2 are supplied in the source package. The finite checks accompanying Figure 2 verify the illustration, not the general theorem.

AI assistance

AI tools assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program. This assistance used OpenAI's ChatGPT. The author directed the development and revision and takes responsibility for the content.

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